

Recreating the Buchla 700 in Pure Data

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1 Introduction

Don Buchla was an American electronic instrument designer and a key figure in the development of electronic music systems. His work significantly shaped experimental music practices during the 1960s and 1970s. Unlike the subtractive synthesis approach popularized by Robert Moog, Buchla’s instruments emphasized alternative methods of sound generation, including complex modulation structures, waveshaping, and non-traditional performance interfaces.

Among Buchla’s later designs was the **Buchla 700**, a digital synthesizer that incorporated frequency modulation (FM) synthesis, waveshaping, and advanced routing possibilities. During the same period, Yamaha’s DX series—particularly the DX7—popularized FM synthesis based on the work of John Chowning. While Yamaha instruments typically used fixed FM operator structures, the Buchla 700 offered a more flexible and experimental architecture.

Despite its powerful capabilities, the Buchla 700 did not achieve widespread commercial success, partly due to its high price and complexity. However, its architecture remains a fascinating example of advanced digital synthesis design.

This project recreates selected elements of the Buchla 700 architecture using **Pure Data (Pd)** as a software platform. The goal is to explore its synthesis structure and provide a functional implementation for learning and experimentation.

2 System Overview

Pure Data was chosen as the development environment due to its accessibility, open-source nature, and extensive community resources. The implementation draws inspiration from both the original Buchla 700 documentation and the ID700 synthesizer, which is based on the voice architecture of the Buchla 700.

Each voice in the system contains up to four oscillators. These oscillators can either be mixed directly to produce the final output or used to modulate one another. The architecture allows complex interactions between oscillators through several modulation indexes.

The signals generated by oscillators are routed through a set of six indexes controlling amplitude and modulation depth. In the original Buchla 700, signals were subsequently processed by two wavetable modules. In this recreation, however, wavetable processing is omitted and signals are routed directly into a morphing stage.

The final output is produced by an equal-power crossfader that blends two oscillator signal paths.

3 Voice Architecture

The original Buchla 700 contains twelve different algorithm configurations that determine the relationships between oscillators and modulation indexes.

Due to the scope of this project, the current implementation includes two primary interaction structures:

- Oscillator-to-oscillator connections enabling **frequency modulation (FM)**.
- Index-to-index connections enabling **timbre modulation**.

These simplified configurations preserve key aspects of the Buchla modulation philosophy while remaining manageable within the Pure Data environment.

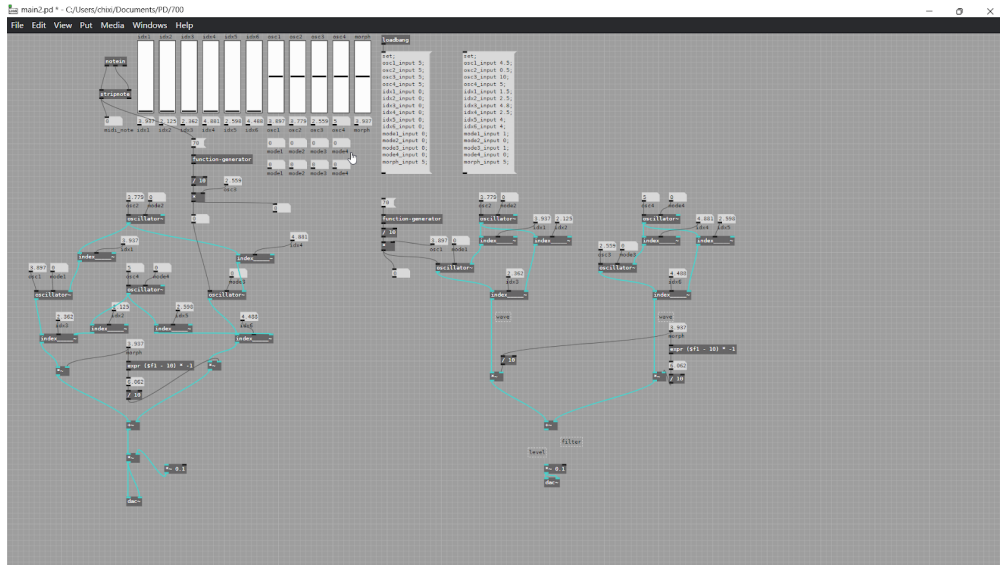


Figure 1: Simplified Buchla 700 oscillator and index architecture

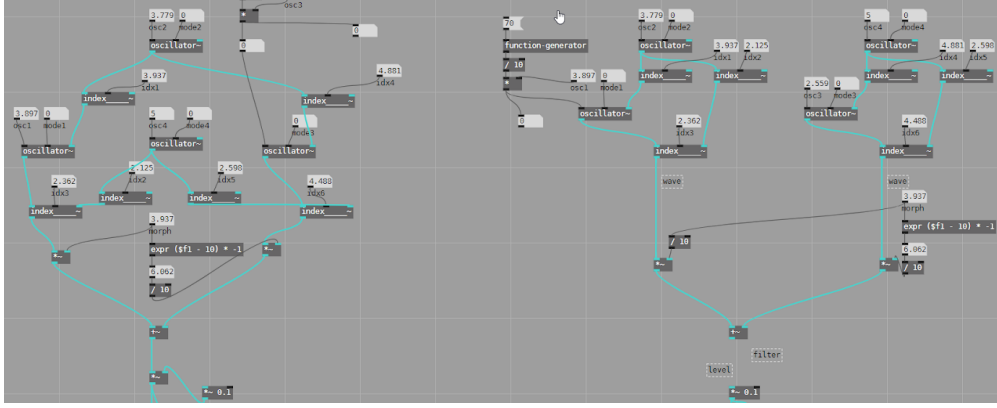


Figure 2: Oscillator routing configuration

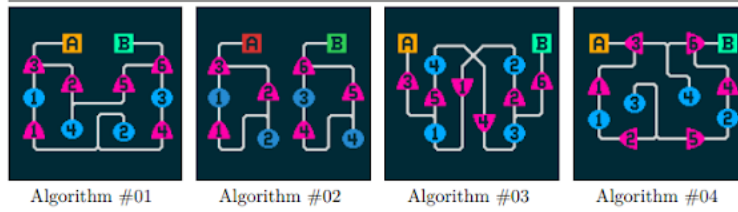


Figure 3: Index interaction structure

4 Oscillator Modes

Each oscillator can operate in three different modes: **Interval**, **Ratio**, and **Frequency**. These modes determine how oscillator frequencies relate to incoming MIDI notes.

4.1 Interval Mode

In Interval Mode, the oscillator frequency is derived from the incoming MIDI note frequency with an adjustable offset. The slider controls pitch deviation within a range of ± 2400 cents, corresponding to $\pm$ two octaves.

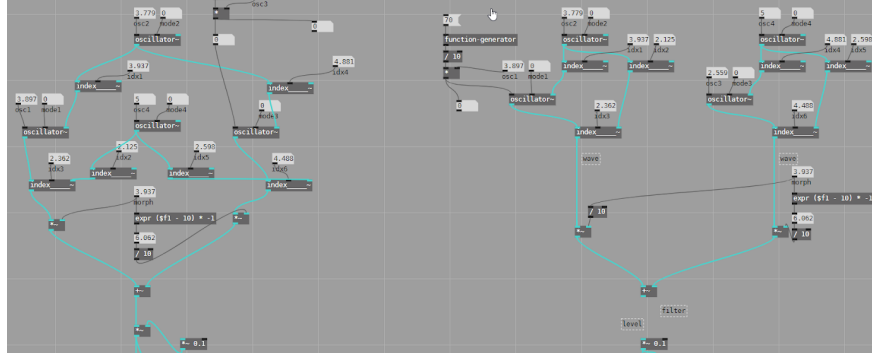


Figure 4: Interval Mode oscillator frequency control

4.2 Ratio Mode

In Ratio Mode, the oscillator frequency is defined as a multiple of the incoming MIDI note frequency. The ratio parameter can be adjusted within a range of 1/16 to 8/1, allowing harmonic or inharmonic relationships between oscillators.

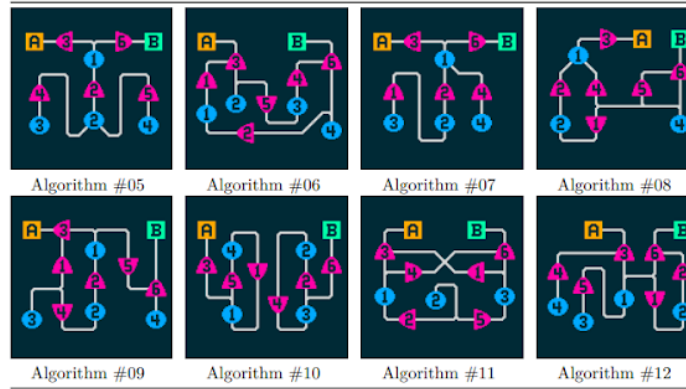


Figure 5: Ratio Mode frequency relationship

4.3 Frequency Mode

In Frequency Mode, the oscillator ignores incoming MIDI pitch information. Instead, the oscillator frequency is directly set by the control parameter within a sub-audio range (0.1 mHz to approximately 16.6 Hz). This mode is primarily used for low-frequency modulation.

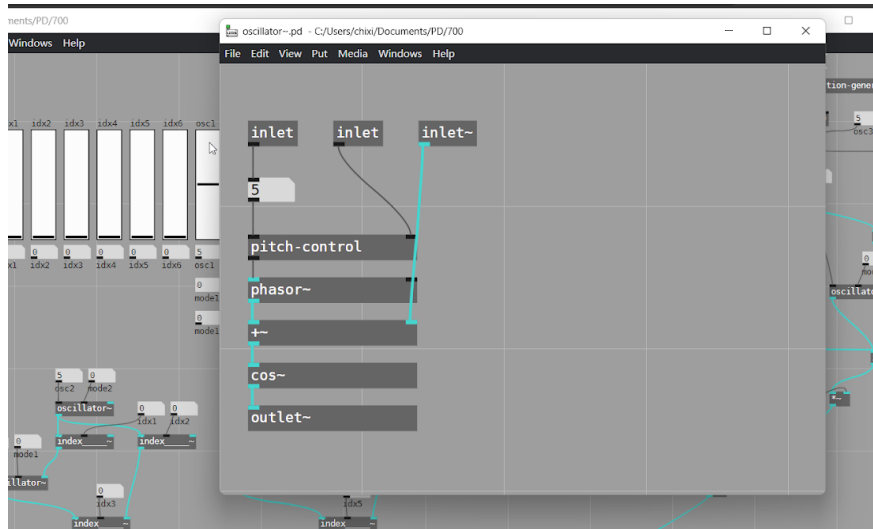


Figure 6: Frequency Mode used for modulation

5 Modulation Indexes

The synthesizer architecture includes several types of modulation indexes that control the interaction between oscillators.

5.1 Amplitude Index

Amplitude indexes control the level of oscillator signals entering the morphing module. These parameters determine the overall signal strength of each oscillator path.

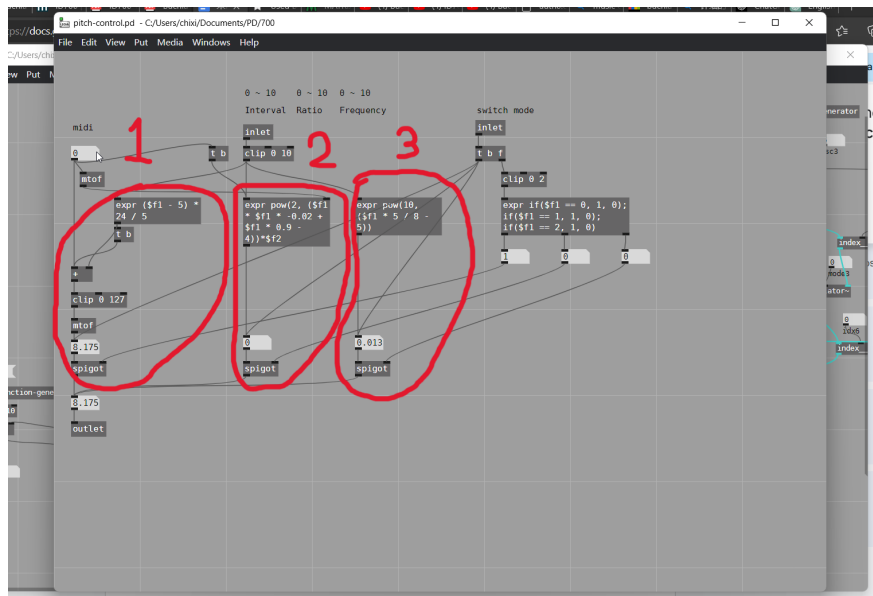


Figure 7: Amplitude index control

5.2 Timbre Index

Timbre indexes connect the output of one oscillator (the modulator) to the input of an amplitude index controlling another oscillator (the carrier). Unlike conventional amplitude modulation, this system multiplies the amplitude index level by a phase-shifted version of the modulator signal.

When the waveform contains a single harmonic component, the result resembles standard amplitude modulation. When complex waveforms are used, additional harmonic interactions emerge, producing richer spectral structures.

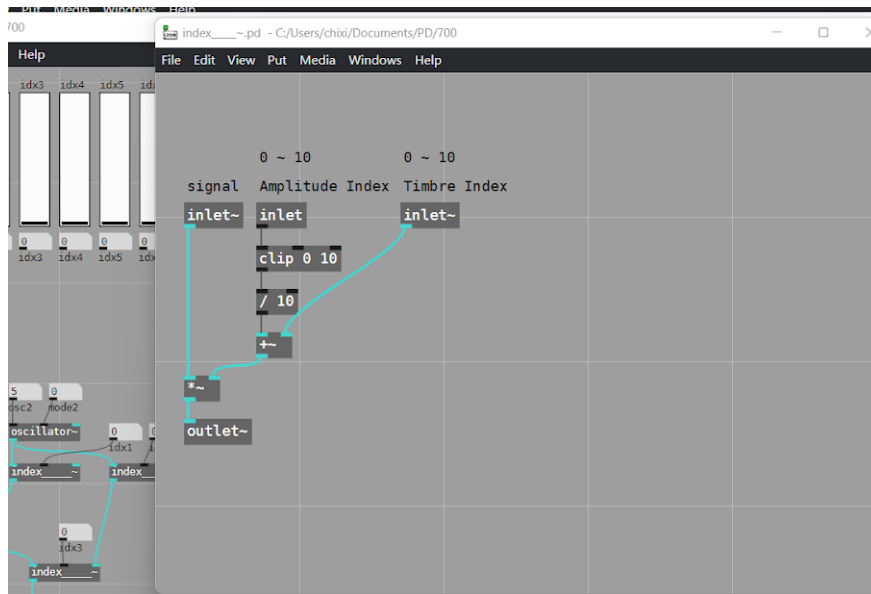


Figure 8: Timbre index routing

5.3 Frequency Index

Frequency indexes enable frequency modulation between oscillators. The output of a modulator oscillator is applied to the frequency input of a carrier oscillator.

The index parameter determines the modulation depth. When the index value is zero, no modulation occurs and the carrier oscillator maintains a stable frequency. As the index increases, the carrier frequency varies according to the modulator waveform.

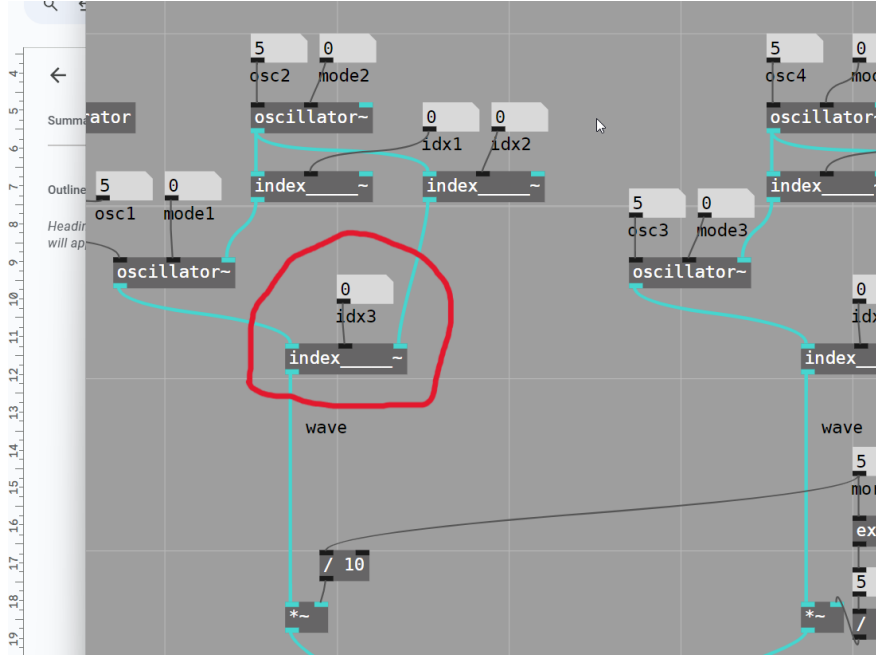


Figure 9: Frequency index modulation path

6 Morphing Module

In the original Buchla 700 architecture, oscillator signals are routed through two wavetable processors before reaching the output stage.

In this implementation, the wavetable stage is omitted. Instead, the signals are routed directly to a **morphing module**. This module performs an equal-power crossfade between two signal paths derived from different oscillator configurations.

The morph parameter allows continuous blending between the two signal structures, enabling smooth transitions between timbral states.

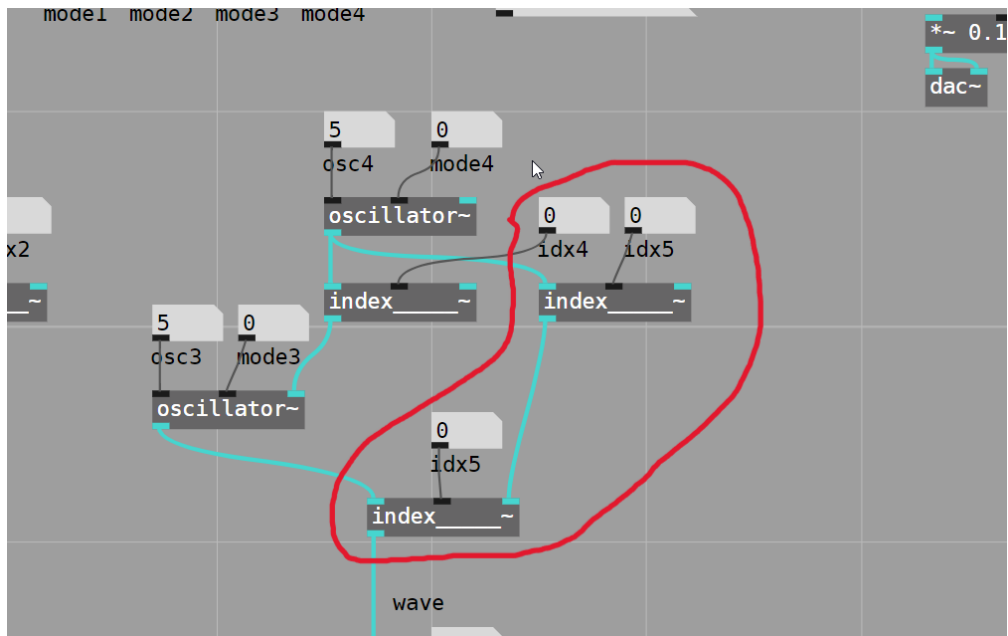


Figure 10: Morph module crossfade



Figure 11: Final output structure

7 Implementation Notes

The current implementation focuses on recreating core synthesis behaviors rather than fully replicating the original hardware architecture. Future improvements may include:

- Implementation of additional oscillator algorithms
- Wavetable processing modules
- Expanded modulation routing
- Voice management and polyphonic control

8 References

References

- [1] Modosc Designs. *ID700 User Manual*. Version 1.30.
- [2] Buchla, Don. *Buchla 700 User Documentation*.